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BAG AND STRAPLESS ATTACHMENT MECHANISM

Abstract

A bag with strapless attachment mechanism is provided. The bag includes a body having at least one interior surface and at least one exterior surface. The bag includes at least two openings each having a closing mechanism for securing the opening in a closed position. Strapless attachment mechanism has a body having at least one magnet disposed in a first end and at least one magnet disposed on a second end distal from the first end. The first end of the body is at least partially disposed on the at least one interior surface of the bag, proximate to one of the openings. A user can wrap the strapless attachment mechanism around an item and attach the at least one magnet of the first end to the at least one magnet of the second, securing the bag to the item of apparel.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of priority of U.S. provisional application No. 63/556,539, filed Feb. 22, 2024, the entire contents of which are herein incorporated by reference.

BACKGROUND OF THE INVENTION
Field of Endeavor
[0002] The present invention relates to bags and attachment mechanisms therefore, and more particularly, to a system including a bag and a strapless attachment mechanism for said bag.
Background of Related Art
[0003] In today's society, people are constantly in motion and have an ever-increasing number of objects they deem essential and always need with their person. This can be problematic when freedom of movement is needed, and comfort is required, for example during exercise or motion-intensive work. Currently, known systems use straps to secure item carriers to a user. However, these systems either provide limited storage space, such as an arm band that can carry a mobile device or provide more storage at the expense of comfort, as strapped systems cause rubbing and chafing proportional to the load they bear.
[0004] As can be seen, there is a need for a system including a bag and a strapless attachment for said bag that can provide increased storage for necessary items and allow for increased comfort.
SUMMARY OF THE INVENTION
[0005] A first aspects of the present invention provides a strapless device for carrying objects.
[0006] The strapless device of the present invention can include a bag having at least a first opening and a second opening; a first closing mechanism, disposed along said first opening; a second closing mechanism disposed along said second opening; a strapless attachment mechanism having a body, at least one interior magnet disposed on the interior of the body at a first end, and at least one exterior magnet disposed on the interior of the body at a second end distal from the first end, wherein at least a portion of the first end is affixed to an interior of the bag proximate to the second opening.
[0007] In another aspect of the present invention, the strapless device can include a plurality of loops, such as belt loops, disposed on an exterior surface thereof. The plurality of loops can be disposed proximate to the second closing mechanism, such as a first loop disposed proximate to a first end of second closing mechanism, and a second loop disposed proximate to a second end of the second closing mechanism.
[0008] In yet another aspect of the present invention, a method of using the strapless is provided. The method of use can include providing a strapless device, comprising: a bag having at least a first opening and a second opening; a first closing mechanism, disposed along said first opening; a second closing mechanism disposed along said second opening; a strapless attachment mechanism having a body, at least one interior magnet disposed on the interior of the body at a first end, and at least one exterior magnet disposed on the interior of the body at a second end distal from the first end, wherein at least a portion of the first end is affixed to an interior of the bag proximate to the second opening; wrapping the body of the strapless attachment mechanism over an apparel item thereby forming an open loop; and closing the open loop by affixing the at least one exterior magnet to the at least one interior magnet.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0009] FIG. **1** is a perspective view of an embodiment of the bag and strapless attachment

mechanism, according to aspects of the present invention;
[0010] FIG. **2** is a side view of an embodiment of the bag and strapless attachment mechanism, according to aspects of the present invention;
[0011] FIG. **3** is a side view of an embodiment of the bag and strapless attachment mechanism showing motion of the strapless attachment mechanism, according to aspects of the present invention;
[0012] FIG. **4** is a rear view of an embodiment of the bag and strapless attachment mechanism, according to aspects of the present invention; and
[0013] FIG. **5** is a rear view of an embodiment of the bag and strapless attachment mechanism showing motion of the strapless attachment mechanism, according to aspects of the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0014] The following detailed description is of the best currently contemplated modes of carrying out exemplary embodiments of the invention. The description is not to be taken in a limiting sense but is made merely for the purpose of illustrating the general principles of the invention, since the scope of the invention is best defined by the appended claims.
[0015] Current systems for efficiently carrying items on a person during exercise, or other movement, require tradeoffs between comfort and capacity. This is due to straps as a primary attachment mechanism. The usage of straps causes rubbing and/or chaffing commensurate with the mass of the items carried. Furthermore, current systems are not designed to be integrated with apparel already in-use because they typically come as a singular unit affixed to a strap. This causes redundancies, as a user who is wearing a belt, is also required to utilize the strap of the bagging system.
[0016] Broadly, an embodiment of the present invention provides a system including a bag and a strapless attachment mechanism, therefore. In a first embodiment of the present invention a user does not sacrifice useful load for comfort as chaffing and rubbing from a strap is eliminated. In a second embodiment of the present invention a system including a bag with belt mounting attachment mechanism is provided. In the second embodiment of the present invention, a bag with loops can be provided for universal attachment to a user's belt, thereby reducing redundancy, and reducing discomfort due to a second strapping system.
[0017] Referring now to FIGS. **1-5**, FIG. **1** illustrates a perspective view of a bag **10** having a strapless attachment mechanism **12**, according to aspects of the present invention. Bag **10** can include a plurality of surfaces such as an exterior surface, and an interior surface, or lining. Bag **10** can have a closing mechanism **18**, for example a zipper, which can be for securing items inside of bag **10**. In embodiments, closing mechanism **18** can be a zipper, Velcro®, hook and loop fasteners, hook and eyes, buttons, buckles, snap fasters, eyelets, or any other known means by which fabric can be secured. In embodiments, strapless attachment mechanism **12** can have at least a portion thereof at least partially affixed to the interior surface of bag **10**. In embodiments, bag **10** can be affixed to an apparel item of a user, such as belt **22**, utilizing strapless attachment mechanism **12**, described further hereinafter.
[0018] FIG. **2** illustrates a side view of bag **10** having strapless attachment mechanism **12**, according to aspects of the present invention. In embodiments, strapless attachment mechanism **12** can include at least one pair of exterior magnets **14**, and at least one pair of interior magnets **16** affixed thereto, or disposed therein. In embodiments, exterior magnets **14** have opposite polarity as interior magnets **16**, such that the magnets can form a magnetic attraction with one another. In embodiments, the portion of strapless attachment mechanism **12** at least partially affixed to the interior surface of bag **10** can include the at least one pair of interior magnets **16**. In embodiments, bag **10** can include an attachment mechanism, such as a strap having one or more fasteners, such as a hook and loop fastener, buckle, button, etc., disposed on the interior thereof proximate to the at least one pair of interior magnets. **16**. In embodiments, the attachment mechanism can be utilized to

secure a device, such as a firearm, phone, flashlight, etc. on the interior of bag **10**. The at least one pair of exterior magnets **14** can be disposed on strapless attachment mechanism **12** distal from the at least one pair of interior magnets **16**. In embodiments, strapless attachment mechanism **12** can be securely enclose an apparel item of a user, such as belt **22**, such that the each of the at least one pair of exterior magnets **14** forms a magnetic attachment to one of the at least one pair of interior magnets **16**. In embodiments, bag **10** can be removed from the apparel item, such as belt **22**, by breaking the magnetic force between exterior magnets **14** and interior magnets **16**, as illustrated in FIG. **3**.

[0019] FIG. **4** illustrates a rear view of bag **10** having strapless attachment mechanism **12**, according to aspects of the present invention. Strapless attachment mechanism **12** can be secured to an interior of bag **10**, as illustrated in FIGS. **2-5**. In embodiments, strapless attachment mechanism **12** can be secured in a main body of bag **10**, or in a separate pocket there from the main body. In embodiments, strapless attachment mechanism **12** can be a fabric material having interior portions configured to accept exterior magnets **14** and interior magnets **16**. In embodiments, interior portion of strapless attachment mechanism can have the at least one pair of interior magnets **16** disposed at a first end, and the at least one pair of exterior magnets **14** disposed at a second end, distal from the first end. It is understood that the at least one pair of exterior magnets **14**, and the at least one pair of interior magnets **16** can be any known magnet of sufficient strength to support a load. For example, the at least one pair of exterior magnets **14**, and the at least one pair of interior magnets **16** can be Neodymium magnets. In embodiments, each magnet can be secured in a separate pocket of strapless attachment mechanism **12**, or alternatively can be affixed to strapless attachment mechanism **12** by fasteners, such as screws, buttons, hook and loop fasteners, etc. It is understood that any known mechanism for affixing magnets to fabric is contemplated without departing from the scope of the present invention.

[0020] In embodiments, strapless attachment mechanism **12** is accessible through an opening on the exterior surface of bag **10**. In embodiments, a second closing mean **20**, for example a zipper, which can be for securing strapless attachment mechanism **12** inside of bag **10**, and for accessing strapless attachment mechanism **12**. In embodiments, closing mechanism **18** can be a zipper, Velcro®, hook and loop fasteners, hook and eyes, buttons, buckles, snap fasters, eyelets, or any other known means by which fabric can be secured. In embodiments, when not in use strapless attachment mechanism **12** can be securable stowed on the interior of bag **10**, or in a separate pocket, as illustrated in FIG. **5**.

[0021] In an alternative embodiment, bag **10** can contain a plurality of loops, such as belt loops disposed on the exterior surface of bag **10**. In embodiments, the plurality loops can be disposed on the portion of the exterior surface having the second closing mechanism **20**. Additionally, the plurality of loops can include at least two belt loops disposed on opposing sides of closing mechanism **20**. In embodiments, a user can place a securing mechanism, such as a belt, through the plurality of loops, which can be then utilized to affix bag **10** to a user.

[0022] Referring now to additional aspects of the invention, an interior of bag **10** can include additional features such as pockets, pouches, hook and loop fasteners, Velcro®, netting, webbing, etc. In embodiments, the additional features of the interior of bag **10** can be utilized to secure items on the interior. For example, one or more pockets, pouches, or attachment interfaces configured to store smaller items, easily damaged or lost in the larger compartment, such as headphones, Tracking Tags, etc. In embodiments, one or more fastening mechanism such as hook and loop fasteners, or Velcro, can be disposed in the interior to secure one or more objects to the interior. Additionally, one or more pieces of netting can be disposed on the interior and can be configured to secure one or more items on the interior of bag **10**.

[0023] Referring now to a method of using bag **10**, a user can remove at least a portion of strapless attachment mechanism **12** from an interior, or pocket, of bag **10** by opening closing mechanism **20**. The user can the wrap strapless attachment mechanism around a structure, such as an apparel item of the user, such that bag **10** becomes securely attached to said structure. In embodiments, the structure can be a shoulder harnesses, dog leash, waist liner of pants, shorts, a collar of a shirt, dress, etc., or other apparel items of the user. It is understood, bag **10** can be affixed to known structures utilizing the wrapping method described above, or alternative can be magnetically affixed in the case that the structure has ferro-magnetic attributes.

[0024] In embodiments, bag **10** and strapless attachment mechanism **12** can be made of any material known in the art. In embodiments, the material of bag **10** and strapless attachment mechanism **12** can include one or more material properties, such as waterproofing, weatherproofing, insulation, reflectivity, bulletproofing, ballistic proofing, puncture resistance, theft and/or tamper resistance, etc. In embodiments, bag **10** can include a tracking device, such as an Radio Frequency ID tag, Apple® AirTag, or other similar wireless tracking devices, installed within bag **10**. Additionally, in embodiments bag **10** can have one or more portions of a clasp, such as a female portion, or male portion, disposed on an exterior thereof and configured to receive one or more modular attachments having the corresponding female or male portion affixed thereto. In embodiments, the one or more portions of clasps can be buckle clasps.

[0025] It should be understood, of course, that the foregoing relates to exemplary embodiments of the invention and that modifications may be made without departing from the spirit and scope of the invention as set forth in the following claims.

Claims

1. A strapless device for carrying objects, comprising: a bag having at least a first opening and a second opening; a first closing mechanism, disposed along said first opening; a second closing mechanism disposed along said second opening; a strapless attachment mechanism having a body, at least one interior magnet disposed on the interior of the body at a first end, and at least one exterior magnet disposed on the interior of the body at a second end distal from the first end, wherein at least a portion of the first end is affixed to an interior of the bag proximate to the second opening.

2. The strapless device for carrying objects of claim 1, wherein the first closing mechanism is one of: a zipper, a hook and loop fastener, at least one button, at least one snap fastener, or at least one magnet.

3. The strapless device for carrying objects of claim 1, wherein the second closing mechanism is one of: a zipper, a hook and loop fastener, at least one button, at least one snap fastener, or at least one magnet.

4. The strapless device for carrying objects of claim 1, wherein the at least one interior magnet is a Neodymium magnet.

5. The strapless device for carrying objects of claim 1, wherein the at least one exterior magnet is a Neodymium magnet.

6. A method of using a strapless device for carrying objects comprising: providing a strapless device, comprising: a bag having at least a first opening and a second opening; a first closing mechanism, disposed along said first opening; a second closing mechanism disposed along said second opening; a strapless attachment mechanism having a body, at least one interior magnet disposed on the interior of the body at a first end, and at least one exterior magnet disposed on the interior of the body at a second end distal from the first end, wherein at least a portion of the first end is affixed to an interior of the bag proximate to the second opening; wrapping the body of the strapless attachment mechanism over an apparel item thereby forming an open loop; and closing the open loop by affixing the at least one exterior magnet to the at least one interior magnet.